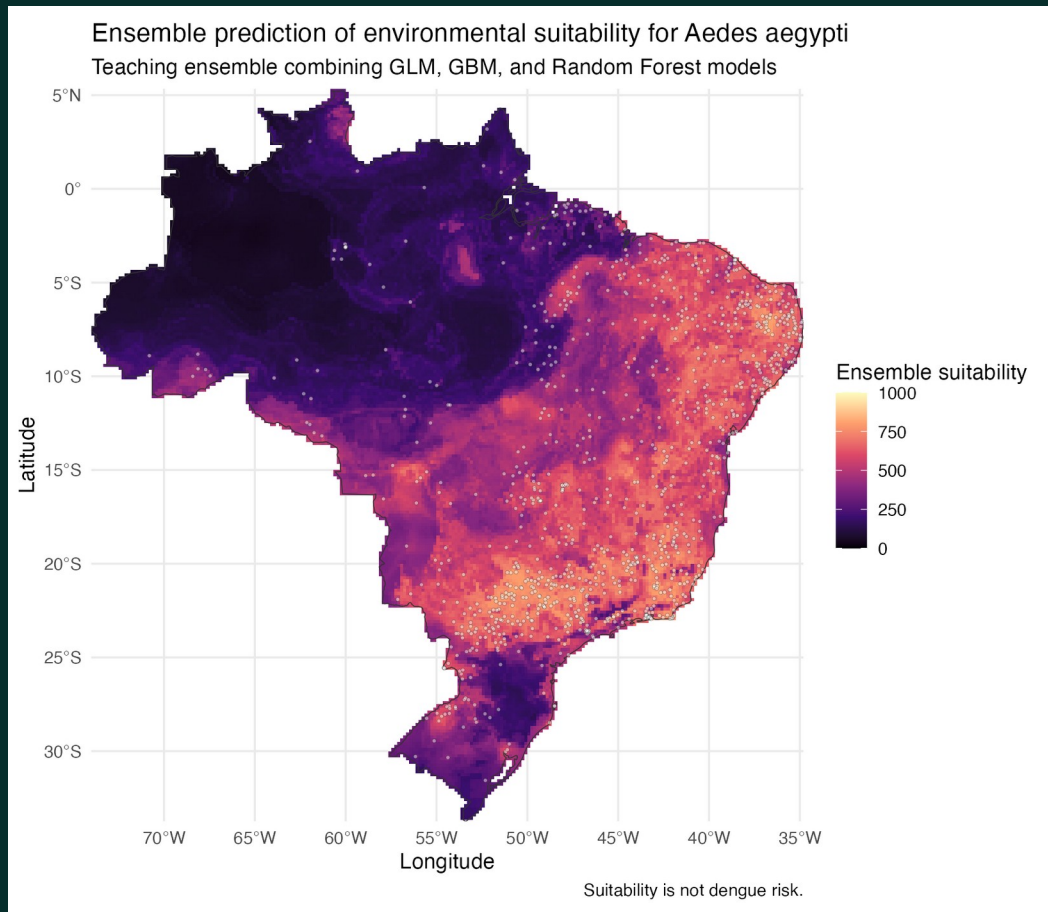


# Species distribution models

Foundations first, then a worked R example using *Aedes aegypti* suitability in Brazil.

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niche | occurrence data | predictors | projection



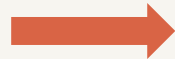
# An SDM links where a species is observed to the environments it occupies

The basic move is simple: compare occurrence locations with environmental predictors, then project that fitted relationship across space.

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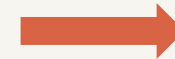
## Records

locations in geographic space



## Environmental relationship

association learned by the model

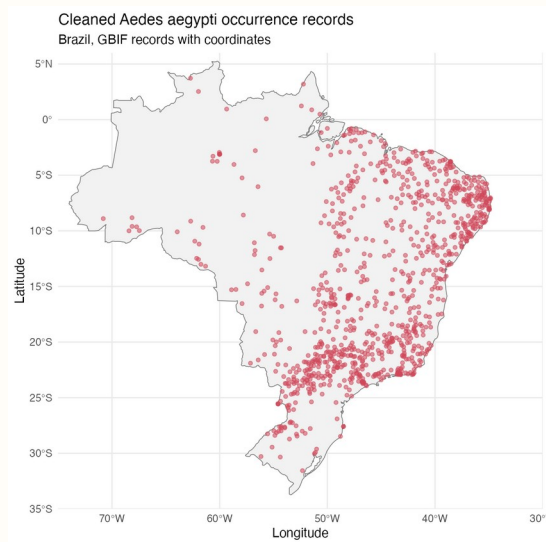


## Projection

a suitability surface

# The output is suitability, not proof of presence

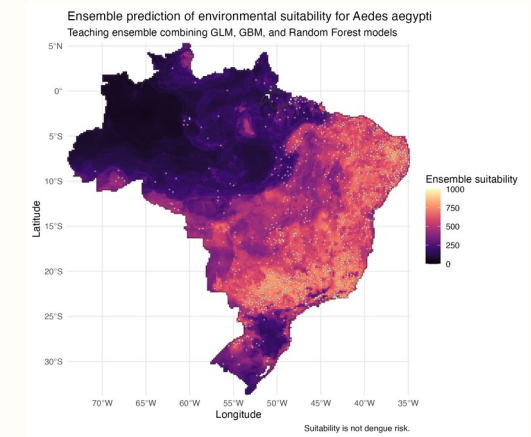
Most SDMs estimate where environments resemble those associated with known records. Biology, dispersal, interactions, and sampling still matter.



geographic observations



**predictor  
values**  
temperature  
rainfall  
seasonality

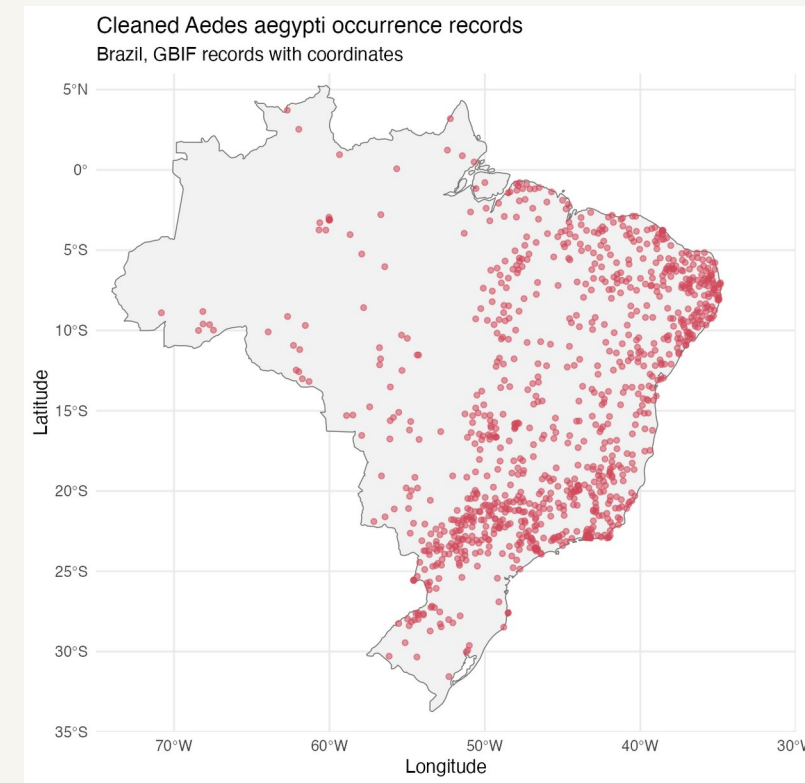


projected suitability

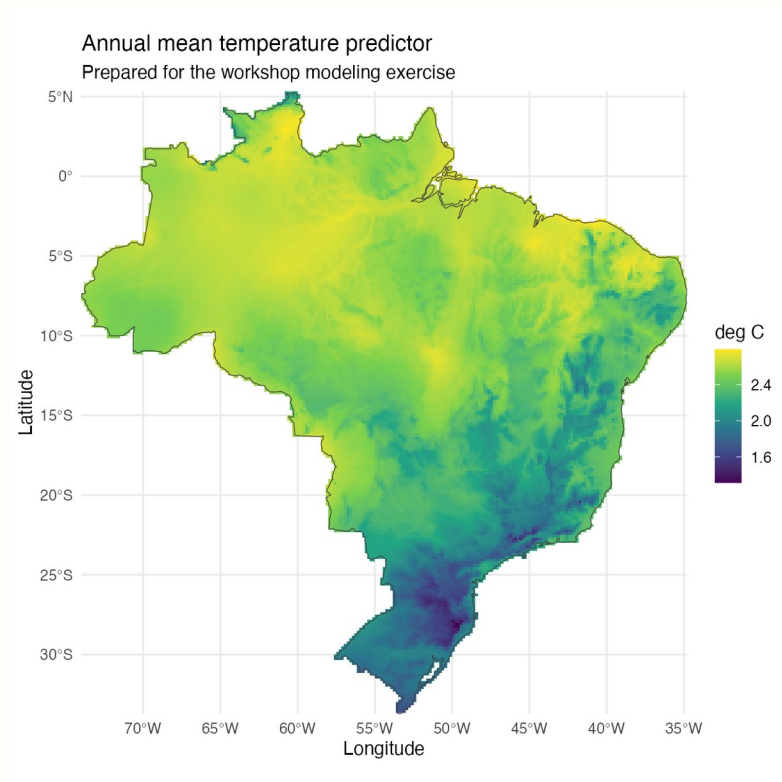
# Case study: *Aedes aegypti* in Brazil

## Why this example?

- The species is epidemiologically relevant.
- GBIF provides reusable occurrence records.
- Brazil gives a national-scale projection problem.
- The result is vector suitability, not dengue risk.



# Predictors define the environmental lens



**We deliberately use four predictors: enough to demonstrate the workflow, few enough to understand the model behaviour.**

annual mean temperature  
temperature seasonality  
annual precipitation  
precipitation seasonality

*Teaching point: fewer variables make the first model easier to inspect.*

# Presence-only data force a modelling choice

Pseudo-absences are comparison points. They are not confirmed places where *Aedes aegypti* is absent.

---

**presence  
records**  
known points



**pseudo-absences**  
background contrast



**biomod2 formatted  
data**  
ready for algorithms

# The three example algorithms bring different biases

The goal is not to crown a winner. It is to see how modelling assumptions affect predictions.

## GLM

**transparent  
statistical trend**

simpler shape

## GBM

**boosted sequence  
of trees**

flexible shape

## RF

**many trees voting  
together**

robust shape

# Evaluation is a stress test, not a certificate

Training data — fitted model — test data

## TSS

threshold-based balance  
of sensitivity and  
specificity

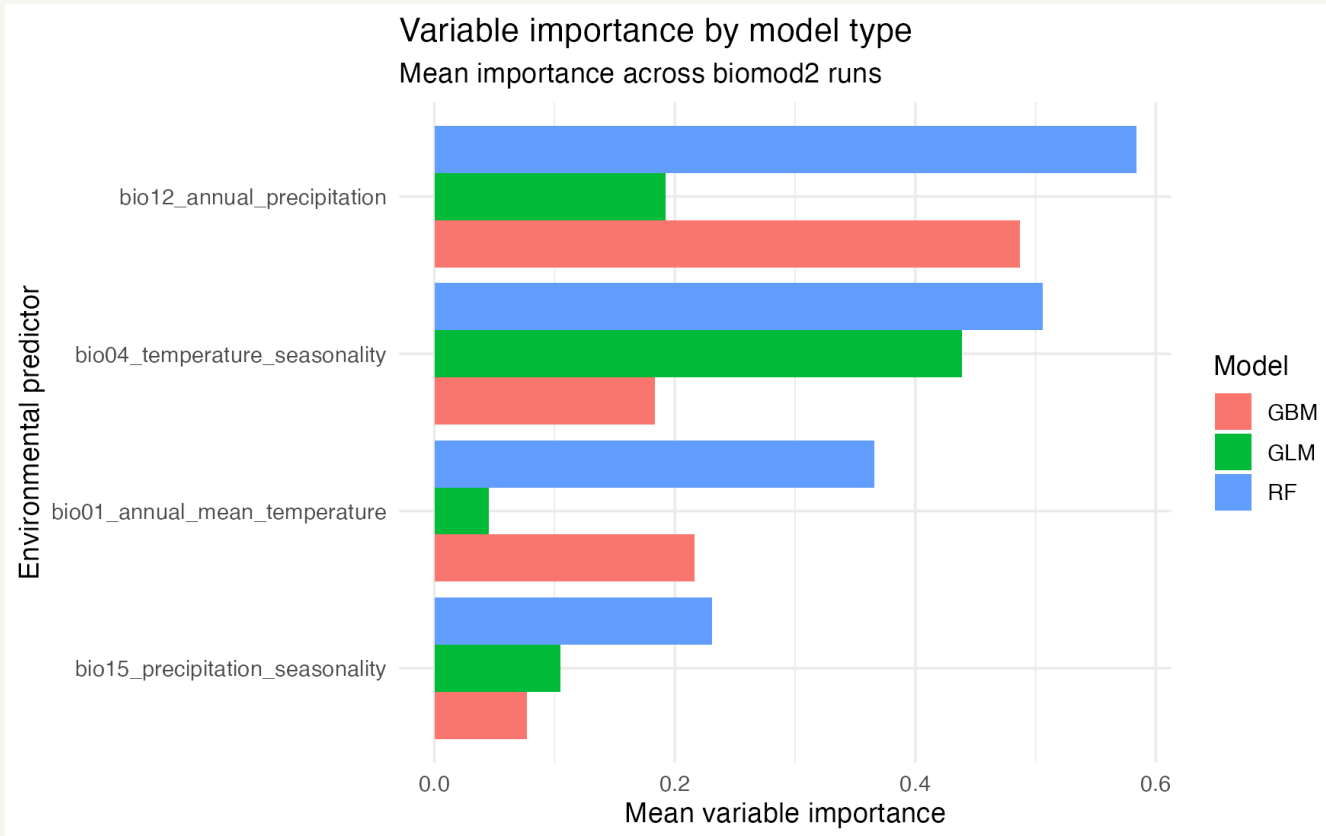
## AUCro

threshold-independent  
ranking of presences above  
absences

**Good scores make a model worth inspecting. They do not make the map true.**



# Variable importance turns outputs



## Read as sensitivity

- Which predictors changed predictions most?
- Do GLM, GBM, and RF agree?
- Could this be ecology, sampling, or correlation among predictors?

# The ensemble is a disciplined summary of disagreement

A mean ensemble reduces the map set; a weighted mean lets better-scoring models contribute more.

GLM

GBM

RF

---

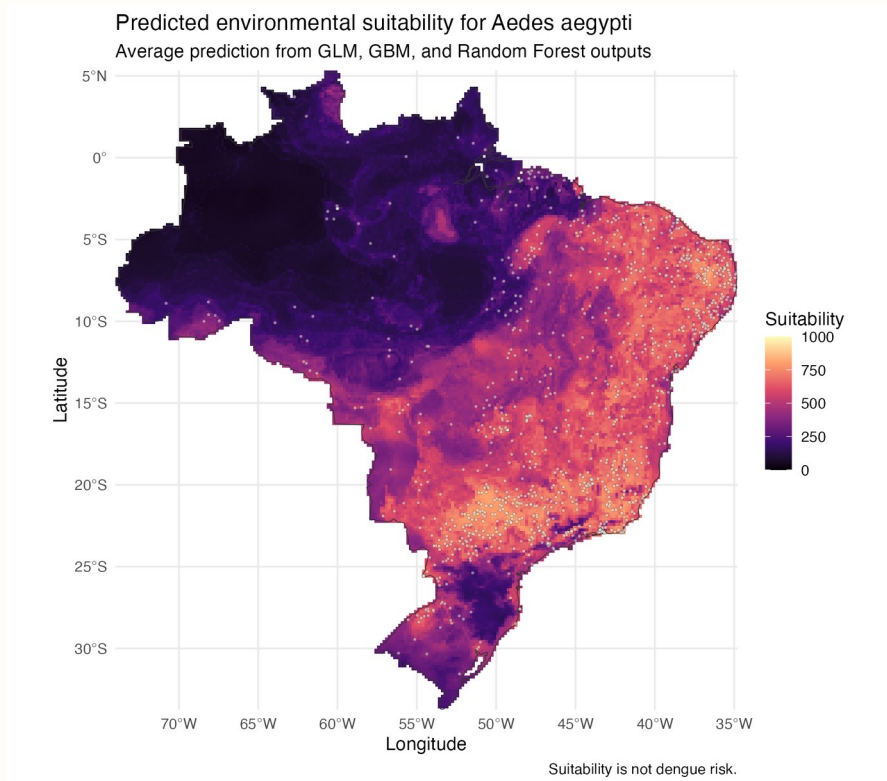
**EMmean**

average prediction

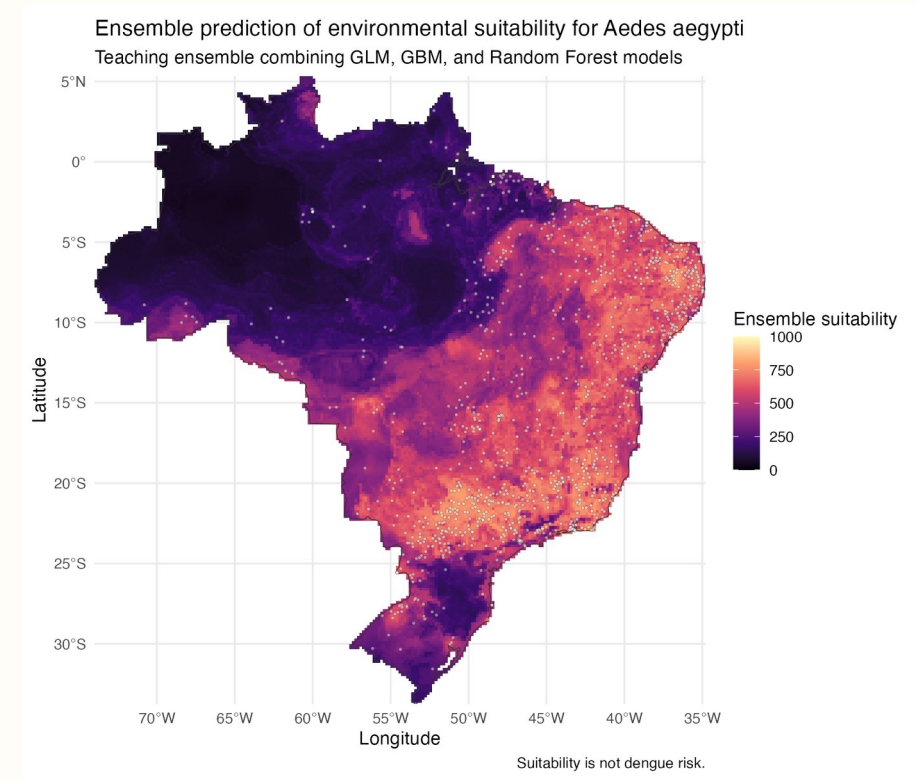
**EMwmean**

performance-weighted  
prediction

# Projection makes model choices visible

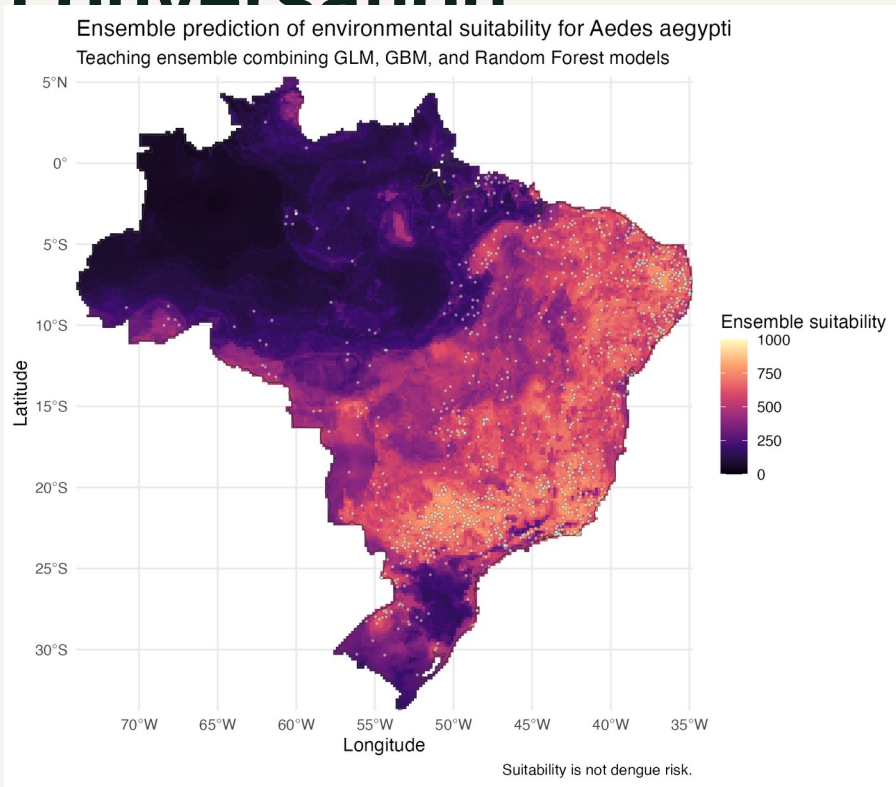


individual model average



ensemble projection

# A good SDM should sharpen the next conversation



**Before acting on a map,  
ask:**

- Where is suitability high or low?
- Where are the data thin?
- Where might sampling bias dominate?
- Which predictor story is plausible?
- What dengue surveillance data would you add?

**Now we build it line by line  
in R.**